

Mechanism-checked error allowance

New audit | 21 September 2026

The previous 100% result survives direct checks of the proposed growth mechanism at the measured powers. Ablations show that correcting the boundary condition accounts for most of the coverage gain. They do not identify the KZ exponent as uniquely responsible.

Ablation: what actually supplies coverage?

Allowance	Pooled coverage	Trial success / CP lower	Mean S/e
Boundary only	0.164	0/3 / 0.000	0.390
Fitted slope + boundary	0.997	2/3 / 0.135	0.209
KZ slope only	0.296	0/3 / 0.000	0.402
KZ slope + boundary	1.000	3/3 / 0.368	0.240
Reference 0 + boundary	1.000	3/3 / 0.368	0.218
Reference 1 + boundary	1.000	3/3 / 0.368	0.260

348 configurations reuse three distinct test trials. Trial success means all configurations for that trial passed. CP values are one-sided 95% nominal binomial lower bounds, not intervals on the pooled rate. Same MLP(8), seed 0 and legacy LOTO b as the prior ablation; no models were retrained.

Does the proposed growth allowance hold?

Law configuration	Endpoint checks	Interval checks	Largest interval ratio
Ceiling 146 mW	4/4	4/4	0.741
Ceiling 192 mW	3/3	3/3	0.696
22-split sweep	29/29	29/29	0.696

Growth is independent of anchor, so 12 repeated anchor checks were removed. The 29 sweep checks reuse three test trials and several intervals; they are not 29 independent observations. Ratios below one support the assumed envelope on measured secants only.

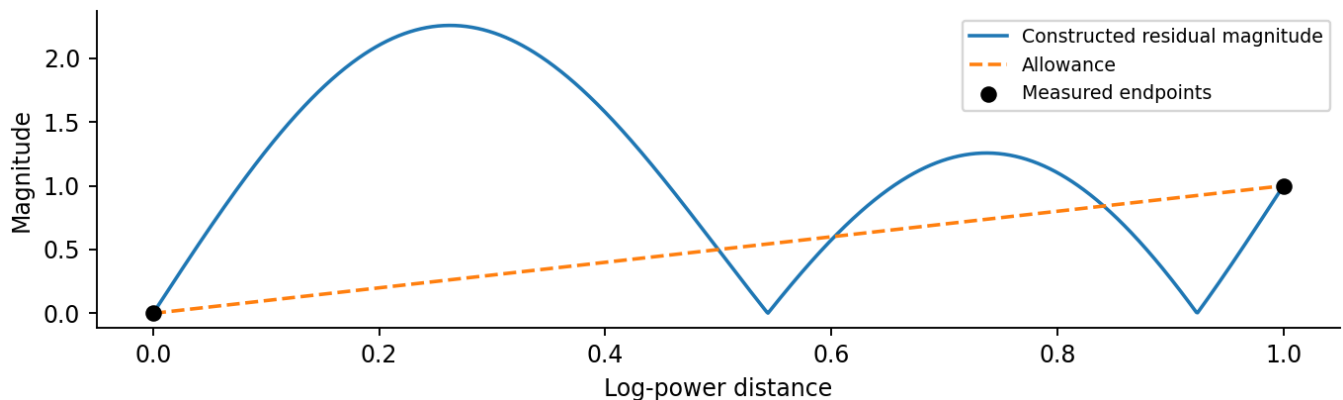
What this supports, and what it does not

The exact decomposition

Let r_0 be the residual at the calibration ceiling, δ the log-power distance, q the fitted law slope and q_{sec} the measured ceiling-to-test slope. The identity $r = r_0 + \delta(q - q_{\text{sec}})$ holds numerically for every audited configuration. The proposed allowance is $\text{norm}(r_0) + k\delta$.

Its margin equals growth slack plus triangle-inequality slack: $[k\delta - \text{norm}(\delta(q - q_{\text{sec}}))] + [\text{norm}(r_0) + \text{norm}(\delta(q - q_{\text{sec}})) - \text{norm}(r)]$. The first term is nonnegative in every measured check, so endpoint coverage does not require vector cancellation to rescue a violated growth budget.

A falsification control: unseen excursions



A deterministic curve $r(\delta) = \delta + 2\sin(2\pi\delta)$ passes both endpoint checks but exceeds the allowance δ by 2 inside the interval. This control shows why finite secant checks cannot prove a continuous derivative envelope. It is a mathematical example, not another physical trial.

Revised claim and next decisive evidence

An explicit calibration-boundary residual plus a slope allowance repairs the failed origin-only envelope on this corpus. All measured secant-growth checks pass. The ablations support the boundary correction, while reference exponents 0, 0.5 and 1 are indistinguishable by coverage. These results establish a mechanism-supported retrospective case study, not 90% population coverage or validation of a unique weak-turbulence exponent.

Freeze the 0.5-reference rule unchanged before new data collection. Obtain denser powers to test for interior excursions and independently acquired held-out trials, ideally on another apparatus. At least 29 independent successes without failure are needed for a one-sided 95% binomial lower bound above 0.90; this is a best-case minimum, not a powered study design.